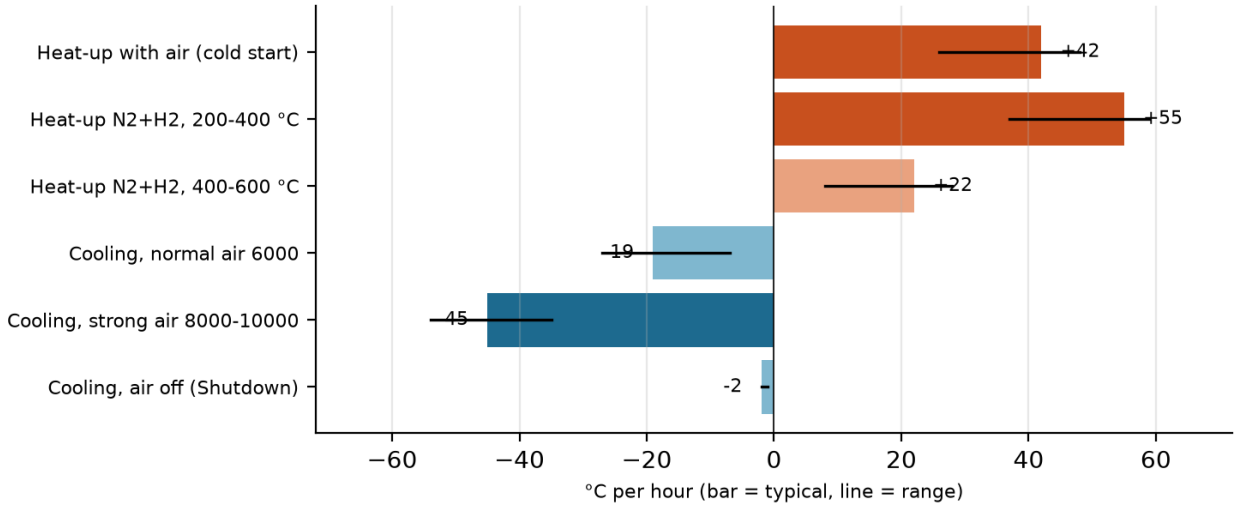
How fast the stack heats and cools

SOFC plant data, 39 logged days (Oct 2025 – Mar 2026). Made 2026-11-03.

Answer: heating runs at about **40–55 °C per hour**. Cooling runs at about **20 °C per hour with normal air** and **45 °C per hour with strong air**. These speeds agree with the delays we calculated earlier. The one exception is "500 °C after 13.5 hours", because that figure also counts time the plant spends waiting.

The slope is how many degrees the stack temperature changes in one hour. We measured it on every heat-up and cool-down run, using the stack temperature and the 8 decided plant stages.



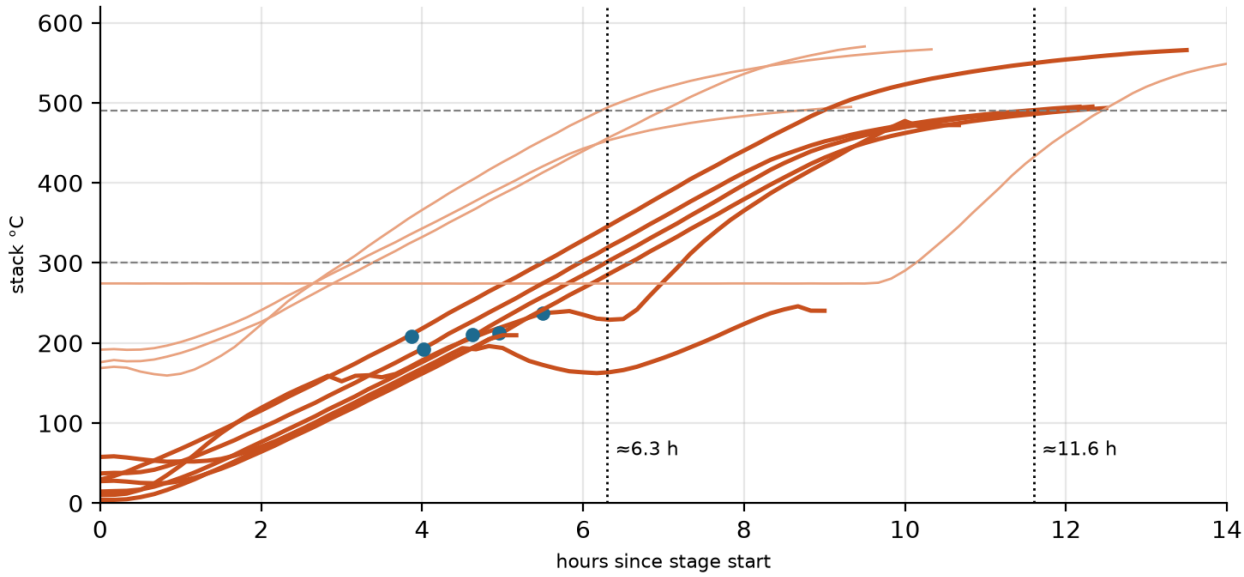
Source: stack temperature, 1-minute data, 12 heat-up and 12 air-flow cooling runs.

Stage	Typical	Range	What it means
Heat-up with air	+42 °C/h	+22 … +48	Room temperature to about 205 °C in about 4½ h. Warm restarts are slower (+9 to +26).
Heat-up with N2+H2	+39 °C/h	+37 … +59	+55 up to 400 °C, then +28, then +16 near the target (about 495 or 570 °C).
Hot standby	≈ 0	–5 … +8	Holding. Target raised: about +18 °C/h. Target lowered: –20 to –28 °C/h.
Air-flow cooling, normal air	–19 °C/h	–7 … –27	About –25 when hot and about –6 below 300 °C.
Air-flow cooling, strong air	–45 °C/h	–35 … –54	About 2½ times faster than normal air.
Shutdown, air off	–2 °C/h	–1 … –2	Only the insulation loses heat. 200 °C is still about 180 °C the next morning.

If you remember one thing: the stack heats at about 40 °C per hour and cools at about 20 °C per hour (45 with strong air).

Heat-up: air first, then N2+H2

From room temperature the stack climbs steadily at 40–55 °C per hour. It slows only when it gets close to its target.



Dark lines = cold starts, light lines = warm restarts, blue dots = gas switches from air to N2+H2.

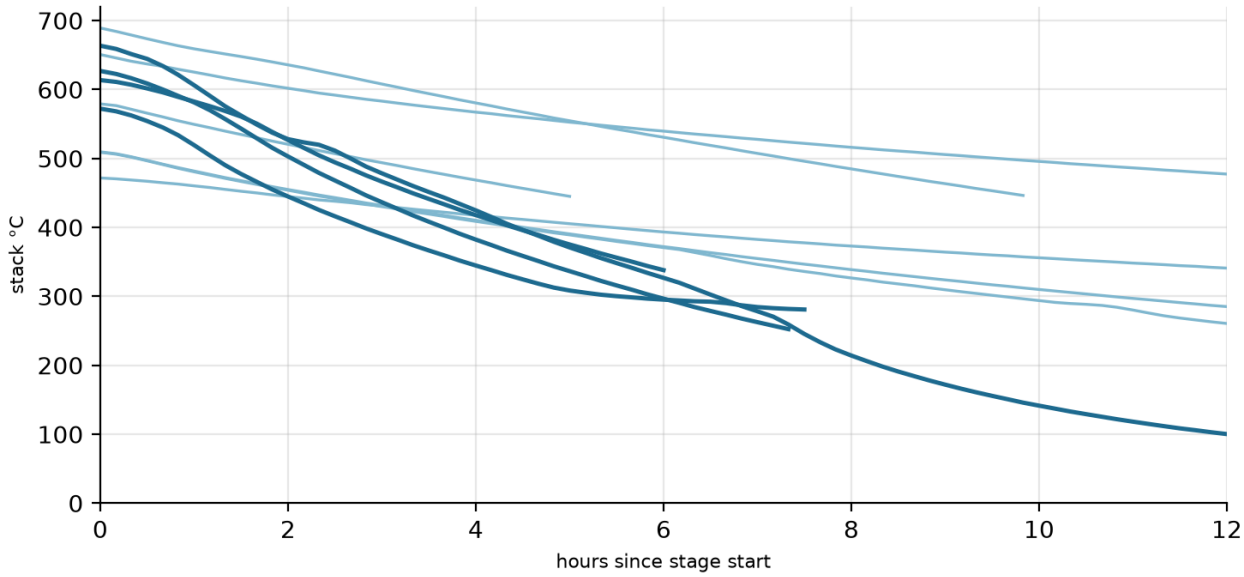
Stack temperature	Heat-up with air	Heat-up with N2+H2
0–100 °C	+38 °C/h	–
100–200 °C	+49 °C/h	–
200–300 °C	≈ 0 (hold before gas)	+54 °C/h
300–400 °C	–	+56 °C/h
400–500 °C	–	+28 °C/h
500–600 °C	–	+16 °C/h

Cold-start milestone	Typical	Range
Stack starts to warm (+2 °C)	28 min	5–66 min
Gas switches to N2+H2 (about 205 °C)	4.6 h	3.9–5.5 h
Stack reaches 300 °C	6.3 h	5.5–7.2 h
Stack reaches 490 °C	11.6 h	9.0–12.2 h

If you remember one thing: a cold start reaches 300 °C after about 6 h and 490 °C after about 11½ h.

Cool-down: the air flow sets the speed

Once the heater is off, air blown through the stack cools it. Strong air (8000–10000 litres per minute) cools about 2½ times faster than the normal 6000.



Dark blue = strong air 8000–10000, light blue = normal air 6000. Each line is one run that started above 400 °C.

Stack temperature	Normal air 6000	Strong air 8000–10000	Air off
600–700 °C	–25	–38	–
500–600 °C	–20	–65	–
400–500 °C	–21	–56	–
300–400 °C	–9	–40	–
200–300 °C	–7	–32	–2
100–200 °C	–6	–25	–2
below 100 °C	–2	–9	≈ 0

All values are in °C per hour. In the first hour normal air takes off about 28 °C and strong air about 50 °C. After 3 h the totals are about 80 and 185 °C.

If you remember one thing: strong air cools at about 45 °C per hour and normal air at about 20. Below 300 °C normal air barely helps.

Do the slopes fit the delays?

Yes. Each delay is either a short wake-up before the stack starts to move, or a long time that should equal distance ÷ slope. We checked both kinds.

Delay calculated earlier	What the slopes give	Verdict
Cold heater on → stack starts warming: 24 min	The first 2 °C rise comes 28 min after heat-up starts (5–66 min).	fits
Cold start → 300 °C: ≈ 6 h	About 200 °C at +42, a short hold, then 100 °C at +55: about 6.3 h. The runs took 5.5–7.2 h.	fits
Cold start → 500 °C: ≈ 13.5 h	490 °C is reached in about 11.6 h. The last 10 °C waits at the ~495 °C hold until the operator raises the target, which adds about 2 h of waiting.	fits, with a note
Hot: target raised → stack moves in 10 min, 90 % in ≈ 3.6 h	The stack climbs at +16 to +18 °C/h above 500 °C, so an 80 °C step takes 4–5 h. The heater outlet ramps at about 60 °C/h, which caps how fast the stack can climb.	fits
Heater cut → stack cools: 11 min (6–16)	In air-flow cooling the first 2 °C drop comes after 7 min (4–16).	fits
Heater output drops ≈ 18 min before the standby → cooling border	A 7–11 min wake-up plus a few minutes at −20 °C/h gives a 2–5 °C drop, about where the border is drawn.	fits
Air level: normal −14, strong −41 to −51 °C/h	Normal −19 (−25 hot, −6 cool), strong −45 (−65 to −25). The old single numbers were averages.	fits, with a note

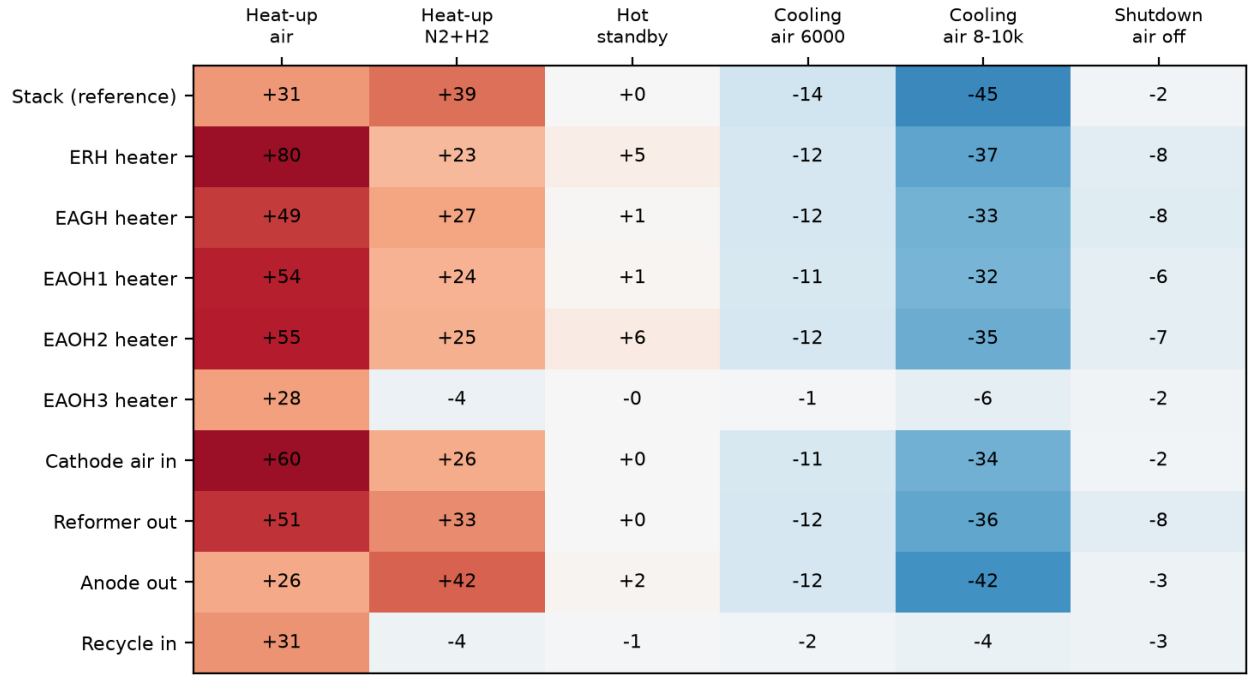
What to change

1. For timing, use "490 °C after about 11.5 h" and count the rest as hold time.
2. Wherever a delay is reused, for example in the synthetic data generator, give the cooling slope per temperature band instead of one number.
3. Check the 2026-03-22 run: 274 °C for 9½ h with strong air, labelled heat-up with air.

If you remember one thing: the delays and the slopes agree. The 13.5 h to 500 °C is waiting time, not slow heating.

Slopes of the other curves

These are the heater and gas-path temperatures drawn next to the stack in the stage figures, measured in the same stages. The heaters move first and fastest. The stack follows 25–30 min later in heat-up and about 5 min later in cooling. That gap is the delay we calculated earlier.



Typical slope in °C per hour (median over runs). Red = heating, blue = cooling.

Curve	Heat-up air begins	Cooling begins, air 6000	Cooling begins, air 8-10k
Stack (reference)	32 min	8 min	6 min
ERH heater	3 min	1 min	2 min
EAGH heater	4 min	3 min	5 min
EAOH1 heater	5 min	2 min	2 min
EAOH2 heater	3 min	2 min	2 min
EAOH3 heater	4 min	3 min	2 min
Cathode air in	3 min	4 min	2 min
Reformer out	8 min	3 min	6 min
Anode out	9 min	13 min	10 min
Recycle in	12 min	5 min	3 min

Minutes until the curve has moved 2 °C (median of runs).

Stage	What the other curves do
Heat-up with air	The heaters jump first: ERH about +210 °C and EAGH/EAOH1/2 +100–135 °C in the first 30 min. Then they climb at about +50 °C/h against +31 for the stack. Cathode air in climbs at +60 °C/h.
Heat-up with N2+H2	At the gas switch EAGH/EAOH1/2 dip 15–40 °C and rise again only after 1½–2 h, then climb at about +25 °C/h against +39 for the stack. EAOH3 and recycle in fall slightly (–4). Anode out (+42) tracks the stack.
Hot standby	All flat (–1 to +6 °C/h).
Air-flow cooling, air 6000	The heaters and gas temperatures all fall at –11 to –12 °C/h, close to the stack (–14). EAOH3 and recycle in barely move (–1 to –2).

Stage	What the other curves do
Air-flow cooling, air 8-10k	About 3 times faster: heaters −33 to −37, anode out −42, stack −45. EAOH3 and recycle in −5 to −6.
Shutdown, air off	The heaters cool 3–4 times faster than the stack (−6 to −8 against −2 °C/h).

If you remember one thing: in every stage the heaters move first and the stack follows. That lag is the delay we calculated earlier.